AgentRoute: Learning-Based Semantic Capability Routing for AI Agents in Mobile Networks

CSE 476/575 Mobile Communication Networks · Term Project Proposal
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Target layer: Network Layer (addressing, routing, forwarding)

We propose AgentRoute, a network-layer scheme in which a packet is addressed to a *capability* instead of a host, and every router learns where to forward it with deep reinforcement learning.

1. Motivation and Problem

Today's Internet routes packets to locations: a host has an IP address, and people reach services through DNS names they already know. AI agents ask a different question: *who can perform this task?* They must find peers by capability, at machine speed, in large and changing populations, and many run on mobile devices whose point of attachment keeps changing. A central (DNS-like) registry becomes a bottleneck and a single point of failure, while flooding does not scale. Recent proposals add agent-specific layers above HTTP [1] or capability discovery services [3], [4]; we instead redesign the network layer itself, so that the address of a destination is what it can do.

2. Proposed Approach (Figure 1)

- **Capability addresses:** each agent encodes what it can do (e.g., "object detection in images") as an embedding vector with a pre-trained sentence encoder; this vector plus an agent ID is its address.
- **Semantic FIB (S-FIB):** instead of IP prefixes, each router keeps aggregated capability vectors per neighbor, filled by bounded-radius capability advertisements.
- **Intent-based forwarding:** a task packet carries an intent embedding; each hop forwards it toward the neighbor whose capabilities best match, until a capable agent is reached.
- **Learned next hop:** a Deep Q-Network at each node selects the next hop from local state (similarity scores, queue load, link quality, handovers); reward = −delay + delivery bonus − drop penalty.
- **Mobility:** when an agent's device hands over between cells, S-FIB entries are updated incrementally instead of re-flooded.

Unlike recent heuristic semantic forwarding (e.g., SPFR [2]), AgentRoute learns its forwarding policy and is evaluated in a mobile cellular/edge topology.

3. Implementation (Code)

All code will be written in Python and included in the final report. A discrete-event simulator (SimPy + NetworkX) models base stations (gNB) with edge nodes, a core network, fixed edge/cloud agents and mobile agents with random-waypoint mobility. Embeddings use sentence-transformers (MiniLM); the DQN policy uses PyTorch, with a GNN-based variant if time permits. A synthetic dataset of about 50 capability classes with paraphrased task requests ensures that intents never match capability text word for word.

4. Evaluation

Baselines: (i) centralized DNS-like registry, (ii) TTL-limited flooding, (iii) DHT (Chord) exact-match lookup on hashed capability names, (iv) greedy semantic forwarding with the same S-FIB but no learning. **Metrics:** discovery success rate, task completion time, hop count, control-message overhead, scalability from 100 to 10,000 agents, and robustness under mobility and node failures.

5. Work Plan and Roles

Work package	Lead
Simulator, topology, mobility model, baselines	Ömer
Embeddings, S-FIB, capability advertisements, greedy semantic forwarding	Bekir Emre
DQN next-hop policy and training	Bekir
Experiments, plots, final report with code	All

References

[1] C. Fleming, L. Muscariello, V. Pandey, R. Kompella, "A Layered Protocol Architecture for the Internet of Agents," arXiv:2511.19699, 2025.
[2] "SPFR: Semantic Potential Field Routing for the Distributed Internet of Agents," arXiv:2608.25396, 2026.
[3] S. Guo et al., "Agent Discovery in Internet of Agents: Challenges and Solutions," arXiv:2511.19113, 2025.
[4] IETF Internet-Draft draft-wang-dmsc-drisac-00, capability registration and semantic discovery for AI agents, Aug. 2026.

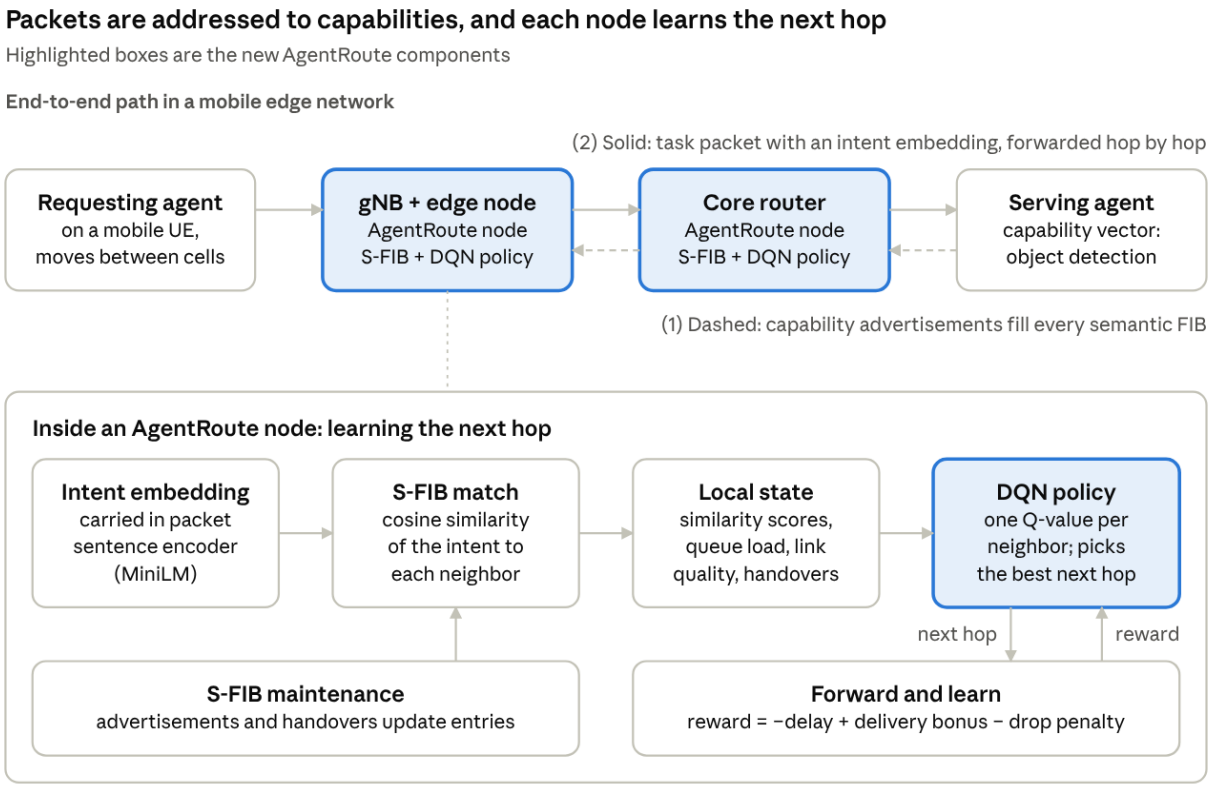


Figure 1. AgentRoute system architecture. Top: capability advertisements (dashed) fill each node's semantic FIB; a task packet carrying an intent embedding (solid) is then forwarded hop by hop to an agent that can serve it. Bottom: inside each AgentRoute node, a DQN policy selects the next hop from the S-FIB match and local network state, and learns from the delay and delivery outcome.